



OPERATIONAL MANUAL

DOOR INTERLOCKING SYSTEM – 2 DOOR

MODEL NO: DLS153-MODBUS

VERSION: 1.0



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Dear Customer,

Thank you for choosing a NEURAL instrument, please read this operating manual carefully before you start using the instrument. This manual will provide you with all the necessary information for the correct use of the instrument as well as a precise idea of its versatility in a wide range of applications.

CAUTION

One three way connector is provided for power connection i.e. Line, Neutral & Earth. (100-220V AC 50/60Hz.)



Do not Switch ON power to the instrument before plugging connectors in to the terminals.



Carefully connect the connectors in the specified terminals if not it may cause damage for the instrument.

Instrument SI. No :

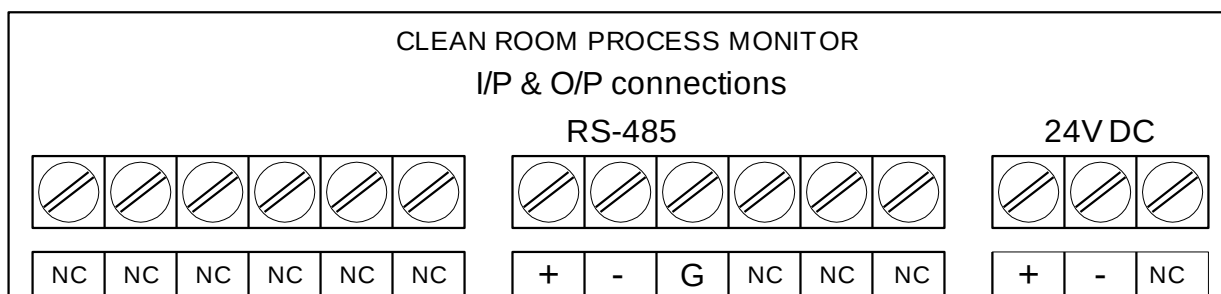
STANDARD PACK

- | | | | |
|-----------------------------|-----|------|--------------------------|
| 1) Instruction Manual | --- | 1 No | ☐ |
| 2) DLS153-MODBUS Instrument | --- | 1 No | ☐ |

Checked By :

Date :

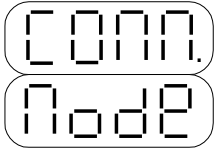
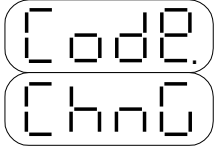
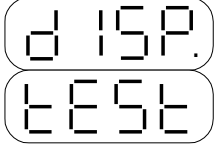
Wiring diagram:



8. Instrument Operation

8.1. Functionality

S.No	Action to perform	Action to be Takes placed	Display
1.	To enter into configuration mode.	Press PRG key, then it will ask for password entry.	
2.	Edit the password to 1111	Use ◀/▶ to edit the value and press OK to conform.	

3.	To go to <i>Communication Parameters sub mode</i> .	Press ▲ key	
4.	To go to <i>Entry Code Change mode</i> .	Press ▲ key	
5.	To go to <i>Display section test mode</i> .	Press ▲ key	
6.	To go to <i>unit selection mode</i> again, or go to previous mode.	Press ▲ key	

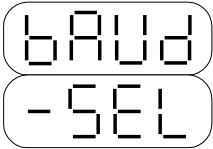
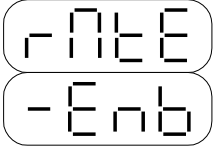
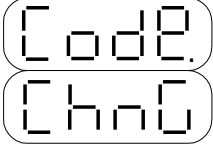
- Press OK key to edit the corresponding programming parameters mode. Then corresponding mode parameters will display and press ESC key to come out from the mode.
- If time out occurs, no key occurs for 30 sec, instrument automatically comes out from configuration mode and enters into measurement mode.
- In the calibration mode this time out is 30 minutes, to stabilize process value.

8.2. Individual mode Description: This section describes about how to edit each programming Parameter mode and brief description about each one.

8.2.1 Communication Parameters Selection sub mode:

Instrument will work as slave module when it connects to MODBUS master. Some of the communication parameters are given to user configurable. These parameters are

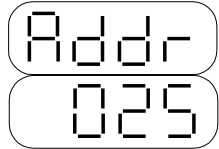
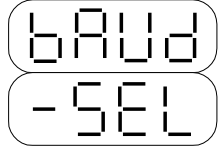
- Instrument address selection
- Baud rate selection

S.No	Description	Action to be Takes placed	Display
1.	After accepting the configuration sub mode By OK key press, when “comm. mode” display, <i>Instrument Address Selection</i> , first programming parameter in the sub mode, will display. Press OK here to edit this mode or to select other parameter use INC/DEC key.	Press OK key	
2.	To go to <i>Baud rate selection</i> parameter. Press OK here to edit this mode or to select other parameter use INC/DEC key.	Press ▲/▼ key	
3.	To go to <i>Remote Programming enable/disable</i> . Press OK here to edit this mode or to select other parameter use INC/DEC key.	Press ▲/▼ key	
4.	To wrap around sub mode parameters press INC/DEC key. To come out from the sub mode press ESC key. Then it will display next parameter Entry code change.	ESC key	

8.2.2 Instrument Address selection:

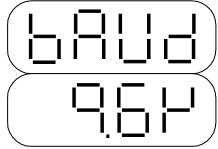
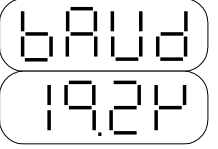
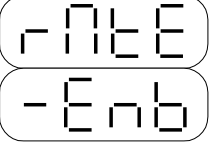
By default instrument address is 1. 247 are the maximum address instrument can support. If user Edit this value more than 247 it will ask for reentry for once. If user enters more than 247 second time also, instrument takes 247 as address and it will go to next parameter selection.

S.No	Description	Action to be Takes placed	Display
1.	<i>Instrument Address selection</i> : After accepting the <i>Instrument address selection by press OK key</i> , it will display present instrument address. By default it is 1.	Press OK key	

2.	Edit the value to required value. Editing digit appears with blinking. To editing digit will select through ◀ key. Press OK key to accept the edited value.	Use ◀/▶/▼ keys	
3.	After press OK key it will display next parameter baud rate parameter selection.	Press OK key	

8.2.3 Baud rate Selection:

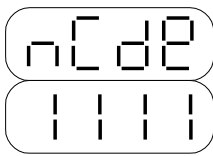
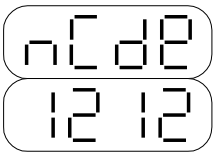
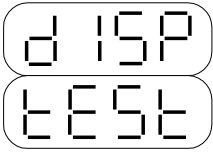
This Instrument supports Three baud rate, 9.6K, 19.2K, 38.4K for RS-485 communication to communicate with any MODBUS master. Default baud rate is 9.6K.

S.No	Description	Action to be Takes placed	Display
1.	<i>Baud Rate:</i> After accepting the <i>Baud rate selection by press OK key</i> , it will display present baud rate for communication. By default it is 9.6Kbps.	Press OK key	
2.	To select required baud rate press INC or DEC key. Press OK key to accept the edited value.	Press ▲/▼ key	
3.	After press OK key it will display next parameter remote programming enable/disable parameter selection.	Press OK key	

8.2.4 Entry Code change:

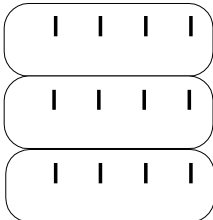
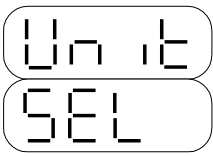
This code should enter to enter in to configuration mode to edit the programming parameters and to do instrument calibration and to do display sections test. Default code is 1111.

This describes how to change the entry code. From next time onwards user should enter new one.

S.No	Description	Action to be Takes placed	Display
1.	<i>New Entry Code:</i> After accepting the <i>Entry Code change</i> by press OK key, it will display present configuration mode entry code. By default it is 1111.	Press OK key	
2.	Edit the value to required value. Editing digit appears with blinking. To editing digit will select through ◀ key. Press OK key to accept the edited value.	Use ◀/▶/▼ keys	
3.	After press OK key it will display first parameter in the mode, display sections test.	Press OK key	

8.2.5 Display Section test:

This is to test display sections proper working. Each segment of 7 segments will display sequentially.

S.No	Description	Action to be Takes placed	Display
1.	<i>Display Sections Test:</i> After accepting the <i>display sections test</i> by press OK key, it will display each segment position.	Press OK key	
2.	Automatically comes out from this test, when completes the two times checking of every segment position or press ESC key. After coming out from this mode it will display first parameter Unit Selection.	Press ESC key	
3.	To come out from the configuration mode Press ESC key. Then it will enter in to measurement mode.		

9. MODBUS Communication:

DLS153 supports MODBUS communication. It works as slave device only. It works on MODBUS serial RTU mode.

9.1. Communication settings:

Mode: RTU mode.

Baud rate: 9600bps (default)

Bits Per Byte: 8-bit

Start bit: 1

Stop bit: 1

Parity: None

Note: DLS supports variable baud rate. User can change the Baud rate through keypad in programming mode. Supported baud rates are 9600, 19200, 38400 bps.

9.2. Memory Mapping:

It supports function codes; listed “Read input registers. Each Register is of length 16 Bit. Signed integer values are stored as standard IEEE- 754 integer format. This format occupies 2 bytes.

9.2.1. Input Registers (3X): Input Registers stores process value of the parameters in integer format.

S.No	Register Address	No of Registers (16 bit)	Parameter Description	Remarks
1.	30300	1	DOOR1 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1
2.	30301	1	DOOR2 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1
3.	30302	1	DOOR3 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1
4.	30303	1	DOOR4 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1
5.	30304	1	DOOR5 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1

6.	30305	1	DOOR6 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1
7	30306	1	DOOR7 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1
8	30307	1	DOOR8 STATUS	This value is stored as Signed Integer Data type Door close: 0 Door open: 1

Appendix:

How to check RS-485 Communication:

1. If a reading from instruments fails, then do the following steps.
2. Switch off the power of all the instruments in the network.
3. Disconnect the instrument to be checked from the network and switch ON the instrument.
4. Check the voltages across + and -, + and G, - and G terminals at the instrument.
5. The voltage across + and - should be greater than or equal to 3.0V
6. Voltage across + and G should be greater than voltage across + and - terminals.
7. Voltage across - and G should be difference of voltage across +/G and +/- terminals.
8. If the steps 5 to 7 are met checked instrument RS-485 section working properly, otherwise remove the instrument and contact factory.
9. Do the same procedure for all the instruments. If all the instruments RS-485 section is OK, check the input lines to the DLS153 network.

Ordering Information**DLS153- MODUS:**

DLS153	X	X	X
Model No	Input type	O/P Type	Future Use
DLS153	MODBUS	1- RS485	X

NOTE:

For Further Information Contact:

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